

SAT Hard Question Set 13

Jack Zeng

July 12, 2026

Novel Prep

Novel Prep

Phase 2 · Hard Question Practice

Overview

1 Hard Question Set 13

Question 1

A circle has center O , and points A and B lie on the circle. The measure of arc AB is 60° , and the length of this arc is 4 inches. What is the circumference, in inches, of the circle?

- A. 4
- B. 8
- C. 16
- D. 24

Answer 1

Correct Answer: D

Question 2

In the xy -plane, the graph of function f , where $y = f(x)$, has exactly 8 x -intercepts. One of these x -intercepts is $(17, 0)$. The rational function g is defined by:

$$g(x) = \frac{f(x)}{x - 17}$$

In the xy -plane, how many x -intercepts does the graph of $y = g(x)$ have?

- A. 5
- B. 6
- C. 7
- D. 8

Answer 2

Correct Answer: C

Question 3

A researcher investigated two species of mites: a predator and its prey. At the start of a week, there was an equal number of the two species. At the end of the week:

- The number of prey had increased by 2700% of the number of prey at the start of the week.
- The number of predators had increased by 180% of the number of predators at the start of the week.

The number of prey at the end of the week was $p\%$ greater than the number of predators at the end of the week. What is the value of p ?

- A. 9
- B. 90
- C. 900
- D. 9000

Answer 3

Correct Answer: C

Question 4

The exponential function f is defined by

$$f(x) = ab^x,$$

where a and b are positive constants. If $f(s) = t$ and $f(s + 1) = t - 0.87t$, where s and t are constants, what is the value of b ?

- A. 0.13
- B. 0.87
- C. 1.13
- D. 1.87

Answer 4

Correct Answer: A

Question 5

A beekeeper's initial observation of the population of a certain bee colony was 1,100 bees. The beekeeper set a goal to increase the population to 2,600 bees. The beekeeper uses a model that predicts the population of this bee colony:

- begins at 1,100
- increases by 120 bees per week in the first two weeks
- then increases by 180 bees per week until the goal is reached

According to this model, at the end of week w after the initial observation, where $w > 2$, which of the following functions gives the predicted number of bees still needed to reach the beekeeper's goal?

- A. $p(w) = 2600 - 180w$
- B. $p(w) = 2480 + 180w$
- C. $p(w) = 1620 - 180w$
- D. $p(w) = -120 + 180w$

Answer 5

Correct Answer: C

Question 6

Given the quadratic equation:

$$7rx^2 - 28sx + 7r = 0$$

In the equation:

- r is a positive constant
- s is a negative constant
- The equation has two solutions

If $\frac{r}{s} > k$, what is the minimum value of k ?

Answer 6

Final Answer: -2

Question 7

A square map has a side length of 55 inches, and 1 inch on the map represents an actual distance of 13 miles. A larger version of the same map is printed as a square with the side length 90% longer than the side length of the previous map. On the larger map, which of the following is closest to the actual distance, in miles, represented by 1 inch?

- A. 1.30
- B. 6.84
- C. 24.70
- D. 49.50

Answer 7

Correct Answer: B

Question 8

In isosceles triangle ABC , sides AB and AC are congruent. Point D divides side $\overline{BC}$ such that the length of $\overline{BD}$ is $\frac{4}{7}$ of the length of $\overline{BC}$. Point E lies on side AB and point F lies on side AC such that when segments $\overline{DE}$ and $\overline{DF}$ are drawn, $\angle BED$ is congruent to $\angle CFD$. If the length of $\overline{BE}$ is 8, what is the length of $\overline{CF}$?

- A. 6
- B. 14
- C. 32
- D. 56

Answer 8

Correct Answer: A

Question 9

$$\frac{\sqrt[3]{x^{17}y^5}}{5x^2\left(\sqrt[8]{y^8}\right)}$$

For all positive values of x and y , the given expression is equivalent to which of the following?

I $\frac{x^{\frac{14}{3}}y^{\frac{5}{3}}}{5xy}$

II $\frac{\sqrt[3]{x^{11}y^2}}{5}$

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

Answer 9

Correct Answer: C

Question 10

The table shows three values of x and their corresponding values of y , where s is a constant. There is a linear relationship between x and y . Which of the following equations represents this relationship?

x	y
$-2s$	24
$-s$	21
s	15

- A. $sx + 3y = 18s$
- B. $3x + sy = 18s$
- C. $3x + sy = 18$
- D. $sx + 3y = 18$

Answer 10

Correct Answer: B

Question 11

The cost of renting a carpet cleaner is \$52 for the first day and \$26 for each additional day. Which of the following functions gives the cost $C(d)$, in dollars, of renting the carpet cleaner for d days, where d is a positive integer?

- A. $C(d) = 26d + 26$
- B. $C(d) = 26d + 52$
- C. $C(d) = 52d - 26$
- D. $C(d) = 52d + 78$

Answer 11

Correct Answer: A

Question 12

The speed of a vehicle is increasing at a rate of 7.3 meters per second squared. What is this rate, in miles per minute squared, rounded to the nearest tenth? (Use 1 mile = 1,609 meters.)

- A. 0.3
- B. 16.3
- C. 195.8
- D. 220.4

Answer 12

Correct Answer: B

Question 13

The table shows three values of x and their corresponding values of $g(x)$, where

$$g(x) = \frac{f(x)}{x + 5}$$

and f is a linear function. What is the y -intercept of the graph of $y = f(x)$ in the xy -plane?

x	$g(x)$
-20	4
-8	0
10	6

- A. $(0, -8)$
- B. $(0, 5)$
- C. $(0, 8)$
- D. $(0, 40)$

Answer 13

Correct Answer: D

Thank You!

We appreciate your time and focus.

Novel Prep